DO NOW — RECALL

Lesson 132 · Hybrid Encryption: Protecting a Connectior

Name: _____ Class: _____ Date: _____

1. Which key verifies a signature produced using an owner's private key?

2. Which memory is typically volatile: RAM or ROM?

3. In a queue, what does FIFO stand for?

4. What is O(n) search complexity associated with?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

The corresponding trusted public key.

2. Model answer

RAM.

3. Model answer

First in, first out.

4. Model answer

Linear/sequential search in the worst case.

By the end, you can…

• Client validates server identity and securely establishes shared key material.

• Both ends obtain secret symmetric traffic keys.

• Bulk application data travels encrypted under symmetric cipher.

Vocabulary — know these terms

session key — Temporary symmetric key for a connection/session.

hybrid encryption — Combines asymmetric key establishment with symmetric data encryption.

certificate — Credential binding identity to a public key, subject to trust validation.

confidentiality — Preventing unauthorised reading of information.

NEW LEARNING

Read, annotate and explain

Authenticate and agree

Client validates server identity and securely establishes shared key material.

Derive session key

Both ends obtain secret symmetric traffic keys.

Encrypt traffic

Bulk application data travels encrypted under symmetric cipher.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Explain why a modern web connection does not encrypt every file byte directly with an asymmetric cipher.

Check server certificate

Predict the first step from the prompt.

Derive symmetric traffic key

Explain the intermediate result or decision.

Encrypt application data

Modern secure connections typically authenticate with public-key credentials, establish shared session keys (often via ephemeral key agreement) and use efficient symmetric encryption for bulk traffic. Identity checks guard against impostors.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Describe why hybrid cryptography uses asymmetric and symmetric methods together. [6]

Question 2

Explain what a session key is and why it is not the same as a website's public key. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate the assertion 'the padlock means any entered data are safe'. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

HTTPS guarantees the website operator is honest.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.